

Weekly Progress Report

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I. Overview

This week, the primary focus was on establishing a rigorous foundation in mathematical statistics, transitioning from advanced probability concepts to large sample asymptotic behaviour and parametric inference frameworks. The core academic focus involved deep-diving into the relationships between multiple random variables, examining the behaviours of sampling distributions derived from normal populations, and mapping out the mathematical boundaries of estimators and hypothesis testing frameworks. Additionally, significant effort was dedicated to integrating these theoretical statistical paradigms with programmatic simulations and learning resources to ensure a robust, applied understanding of data modelling principles.

II. Achievements

1. Advanced Probability & Sampling Foundations

- **Moments and Generating Functions:** Developed a thorough understanding of the moments of a distribution function, calculating expectations of higher-order powers, and mastering the utility of Moment Generating Functions (MGFs) and Probability Generating Functions (PGFs) to characterize distributions completely.
- **Some Moment Inequalities:** Studied and applied fundamental probability bounds and inequalities (such as Markov's inequality and Chebyshev's inequality) to establish upper bounds on tail probabilities.
- **Multiple Random Variables:** Explored joint, marginal, and conditional probability distributions for vectors of multiple random variables, analysing how individual variables interact within a joint space.
- **Independence and Covariance:** Mastered the mathematical criteria for the independence of random variables, alongside computing covariance and correlation coefficients to quantify linear dependencies.
- **Conditional Expectation:** Studied conditional expectations and conditional variances, treating them as random variables themselves, and utilized the Law of Total Expectation.

- **Order Statistics:** Learned the distribution theory behind order statistics ($X_{(1)}$, $X_{(2)}$, $X_{(n)}$), deriving the probability density functions for sample extremes (minimum and maximum) and intermediate ranks.
- **Sampling Distributions from Normal Populations:** Formally derived the exact sampling distributions of $(\bar{X}, S^2)$ under normal population assumptions, including a deep-dive into the properties and applications
- **2. Asymptotic & Large Sample Theory**
- **Modes of Convergence:** Explored the strict hierarchy of mathematical convergence in statistical theory, distinguishing clearly between convergence in probability, convergence in distribution, and almost sure convergence.
- **Laws of Large Numbers:** Evaluated the mechanics and implications of both the Weak Law of Large Numbers (WLLN) and the Strong Law of Large Numbers (SLLN), establishing how sample means behave as sample size n approaches infinity.
- **Limiting Moment Generating Functions:** Utilized limiting forms of MGFs to prove asymptotic properties and trace the convergence paths of sequences of random variables.
- **The Central Limit Theorem (CLT):** Mastered the Lindeberg-Lévy formulation of the CLT, understanding why independent, identically distributed variables aggregate into a normal distribution asymptotically, regardless of the underlying parent distribution.

3. Parametric Point Estimation & Hypothesis Testing

- **Properties of Estimators:** Studied the fundamental criteria for evaluating point estimators: Sufficiency (Fisher-Neyman Factorization Theorem), Completeness, and Unbiasedness.
- **Neyman-Pearson Fundamental Lemma:** Evaluated the construction of Uniformly Most Powerful (UMP) tests for simple hypotheses using the Neyman-Pearson lemma to optimize test power at a fixed significance level α .
- **Likelihood Ratio Frameworks:** Explored Generalized Likelihood Ratio Tests (GLRT) for complex, composite testing scenarios alongside Monotone Likelihood Ratio (MLR) families.
- **Advanced Testing Procedures:** Introduced the operational concepts behind Bayes and Minimax testing procedures, evaluating decision-theoretic approaches to classical hypothesis testing.

III. Challenges

1. Asymptotic Convergence Proofs

- **Theoretical Rigor:** Encountered significant challenges when working through the real-analysis-based proofs required to separate convergence in probability from almost sure convergence. The mathematical boundaries governing the Strong Law of Large Numbers proved conceptually difficult to visualize purely from textbook definitions.
- **Mitigation Strategy:** Seeking academic guidance and working through additional counterexamples to better comprehend cases where weak convergence holds true but strong convergence fails.

2. Hypothesis Testing Boundaries and Optimality

- **Operational Rules:** Faced complexity when trying to distinguish the exact conditions under which a Locally Most Powerful Test should be preferred over a Generalized Likelihood Ratio Test.
- **Decision Criteria:** Integrating classical Neyman-Pearson frameworks with decision-theoretic approaches (like Minimax and Bayes risk minimization) presented a steep learning curve regarding how loss functions alter optimal decision boundaries.

IV. Learning Resources

1. Primary Textbooks and Documentation

- **Core Reference Text:** Thoroughly reviewed textbook chapters 3, 4, 7, 8, 9, and 10, utilizing detailed chapter proofs and end-of-chapter problem sets as a primary framework for troubleshooting complex equations.
- **Academic Lectures:** Attended supplementary recorded lectures and online tutorials focused on advanced mathematical statistics to break down the factorization theorem for sufficiency.

2. Computational & Practical Application Tools

- **Statistical Simulations:** Utilized Python and specialized statistical packages to build empirical simulations of the Central Limit Theorem. Programmatically sampled from skewed distributions to visualize how rapidly the sampling distribution of the mean approaches' normality as n scales.
- **Peer Frameworks:** Participated in statistical discussion forums to review derivations of order statistics and compare notes on calculating joint distribution bounds.

V. Next Week's Goals

1. Deepening Parametric Testing & Exact Distribution Mechanics

- **Resolve Testing Gaps:** Dedicate focused study time to mastering the mathematical mechanics of Bayes and Minimax procedures.
- **Multivariate Expansion:** Extend current knowledge of bivariate normal distributions to fully multivariate normal distributions, focusing on quadratic forms and their associated distribution theory.
- **Collaborative Review:** Engage in study groups with peers to review edge cases in composite hypothesis testing and discuss proof methodologies for UMP tests.

2. Applied Statistical Modelling

- **Algorithm Implementation:** Implement maximum likelihood estimation (MLE) and method of moments (MoM) estimators programmatically within computational data projects to move from theoretical equations to applied algorithms.
- **Feedback Integration:** Submit derived problem sets regarding exact sampling distributions to mentors/instructors for targeted feedback to ensure absolute analytical accuracy moving forward.

VI. Additional Comments

This week provided an exceptional, albeit challenging, bridge between basic probability calculus and the broader architecture of statistical inference. Moving from the static behaviour of individual random variables to the dynamic, asymptotic behaviour of sample sizes approaching infinity highlights the immense power of the Central Limit Theorem in real-world data science. While the mathematical rigor required for uniformly most powerful tests and asymptotic proofs demands deep concentration, mastering these principles is directly translating into a clearer understanding of how statistical models operate under the hood